Lottery 1 : win 2M€ p: 1%

~~~ 2 : win 10M€ p: 0.1%, 2M€ p: 0.8%, 0€ p: 0.01€

~~~ 3 : win 2M€ p: 0.11, 0€ p: 0.89

~~~ 4 : win 10M€ p: 0.1, 0€ p: 0.9

$$(1)\begin{cases} L_1 : U(2€) \;\bullet \\ L_2 : U(10€)\cdot 0.1 + U(2€)\cdot 0.89 + U(0€)\cdot 0.01 \;\bullet \end{cases}$$

L_1 better L_2 $\Rightarrow$ $\bullet > \bullet$

$$L_4 : U(10)\cdot 0.1 + U(0)\cdot 0.9 > L_3 : U(2)\cdot 0{,}11 + U(0)\cdot 0{,}89$$

(1)

$$U(2€) - 0{,}89\cdot U(2€) > U(10€)\cdot 0{,}1 + U(0€)\cdot 0{,}01$$

$$0{,}11\cdot U(2€) > U(10€)\cdot 0{,}1 + U(0€)\cdot 0{,}01$$

$$\begin{cases} \underline{0{,}11\cdot U(2€) + U(0€)\cdot 0{,}89} > \underline{U(10€)\cdot 0{,}1 + U(0€)\cdot 0{,}9} \\ \underline{U(10€)\cdot 0{,}1 + U(0€)\cdot 0{,}9} > \underline{U(2€)\cdot 0{,}11 + U(0€)\cdot 0{,}89} \end{cases}$$

→ impossible

$$\begin{array}{cc|c|c|c} & & P_2 & & \\ & & H & T & \\ \hline P_1 & H & 1,-1 & -1,1 & p \\ & T & -1,1 & 1,-1 & 1-p \\ \hline & & 0.5 & 0.5 & \\ & & 0.3 & 0.7 & \end{array}$$

Payoff P_2 :

$$0.5p + (1-p)0.5 - 0.5(1-p) - 0.5p = 0$$

$\Rightarrow$

$$u_1(\text{win}) = 0.3p + 0.7(1-p)$$

$$U_1(\text{Lose}) = 0.7p + 0.3(1-p)$$

$$\max_p \; u_1(\text{win}) - u_1(\text{Lose})$$
~~~

$0.3p + 0.7(1-p) - 0.7p - 0.3(1-p) =$

$0.3p + 0.7 - 0.7p - 0.7p - 0.3 + 0.3p = -0.8p + 0.4$

$\max_p (-0.8p + 0.4) \to p = 0 \Rightarrow 0.4$

P_2

P_1	B	S	
B	2,1	0,0	$p = 2/3$
S	0,0	1,2	$(1-p) = 1/3$
	$q = 1/3$	$(1-q) = 2/3$	

$U_2(B) = p$, $U_2(S) = 2(1-p)$

$U_2(B) = U_2(S) \to$

$p = 2(1-p)$

$p = 2 - 2p$

$3p = 2 \to p = 2/3$

$U_1(B) = U_1(S)$

$2q = 1 - q$

$3q = 1 \to q = 1/3$

	B	S
B	$\frac{2}{9}$	$\frac{4}{9}$
S	$\frac{1}{9}$	$\frac{2}{9}$

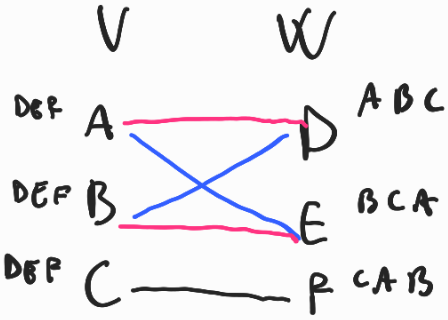
$2 \cdot \frac{2}{9} + \frac{2}{9}$ = payoff for P_1

Stable Matching

2 sets same sizes (males, females)

Players: the members of both sets

Actions: choose a member of the other set



MALES:

A	42315
B	53214
C	25143
D	52431
E	41235

FEMALES:

1	DBECA
2	BADCE
3	ACEBD
4	DACBE
5	BEACD